Chemistry 3A – Fall 2025 Thursday Version**: Section 25811**

Name __Red Hot Chem Student_____ ID __10101010___

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

_**d**__ $V = k \times n$
_**e**__ $PV = nRT$
_**c**_ $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
_**a**_ $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm or 760 torr

(b) What is standard temperature in Celsius (°C)? (1 pt)

0° C

(c) Convert that standard temperature into Kelvin (K). (1 pt)

K = C + 273.15 or 273 = 0 + 273.15 K or 0 + 273 K = 273.15 K or 273 K

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of $\mathbf{2.80\ L}$ at a **temperature** of $\mathbf{22^\circ C}$. The gas is heated at constant pressure until its new volume reaches $\mathbf{3.60\ L}$. What is the *new* **temperature** of the gas in **Celsius (**°C**)**? (Are intermediate conversions necessary?)

1. Convert T_1 to Kelvin:
   $22 + 273 = 295\ \text{K}$
2. Formula: $\frac{V_1}{T_1} = \frac{V_2}{T_2} \Longrightarrow T_2 = V_2 \times \frac{T_1}{V_1}$
3. Calculation:
   $T_2 = 3.60\ \text{L} \times \frac{295\ \text{K}}{2.80\ \text{L}} = 379.28\ \text{K}$
4. Convert back to °C: $379.28 - 273 = 106.28^\circ\text{C} \rightarrow \mathbf{106}^\circ\text{C}$

Rubric (Total 5 pts):

- **5 pts:** Correct setup, Kelvin conversions, algebra, final unit (°C), and correct sig figs (106°C).
- **4 pts:** Correct calculation and units, but forgot the final conversion back to °C (left as 379 K).
- **3 pts:** Correct formula/algebra, but temperature conversion skipped (used 22°C directly).
- **1–2 pts:** Identified variables (V_1, T_1, V_2) and selected the correct formula, but made calculation or algebraic errors.
- **0 pts:** Blank or incorrect approach

4. Kinetic Molecular Theory (3 pts)

According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

Rubric:

- 3 pts: States clearly that collisions are **completely elastic** and **no energy is lost**.
- 2 pts: Mentions energy is conserved or bounces without slowing down.
- 1 pt: General vague statement about moving fast.
- 0 pts: Incorrect statement.

5. Ideal Gas Law Calculation (5 pts)

Calculate the **pressure** (in atmospheres) exerted by $\mathbf{0.150}$ **moles** of carbon dioxide gas (CO_2) confined in a $\mathbf{2.50}$ **L** container at a **temperature** of $\mathbf{310}$ **K**. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

• **Formula:** $PV = nRT \Rightarrow P = \frac{nRT}{V}$ • **Calculation:** $P = \frac{(0.150\ \text{mol})\left(0.08205 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(310\ \text{K})}{2.50\ \text{L}}$ $= 1.53\ \text{atm}$	**Rubric (Total 5 pts):** • **5 pts:** Completely correct setup, correct R usage, proper algebraic isolation of P, and correct answer (1.61 atm). • **4 pts:** Minor arithmetic or rounding slip, but the methodology is clean. • **3 pts:** Correct formula and inputs, but algebraic rearrangement error (e.g., multiplied by volume instead of dividing). • **1–2 pts:** Identified n, V, T values properly, but stalled or used the wrong gas constant. • **0 pts:** Blank.

6. Partial Pressures and Mole Fraction (8 pts)

A gaseous mixture in a sealed container consists of $\mathbf{0.300}$ moles of helium gas (He) and $\mathbf{0.200}$ moles of neon gas (Ne). The total pressure of the gas mixture is measured to be $\mathbf{1.50}$ atm.

(a) What is the total number of moles ($\boldsymbol{n}_{\text{tot}}$) of gas in the container? *(1 pt)*

Total Moles: $n_{\text{tot}} = 0.300\ \text{mol} + 0.200\ \text{mol} = \mathbf{0.500}\ \text{mol}$

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

Mole Fraction: $\chi_{\text{Ne}} = \frac{n_{\text{Ne}}}{n_{\text{tot}}} = \frac{0.200\ \text{mol}}{0.500\ \text{mol}} = \mathbf{0.400}$

(c) Calculate the partial pressure ($\boldsymbol{P}_{\text{Ne}}$) of neon gas in the mixture. *(3 pts)*

Partial Pressure: $P_{\text{Ne}} = \chi_{\text{Ne}} \times P_{\text{tot}} = 0.400 \times 1.50\ \text{atm} = \mathbf{0.600}\ \text{atm}$

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

The mole fraction of neon will **decrease**. *Explanation:* Adding more helium increases the total moles of gas (n_{tot}) while neon's moles (n_{Ne}) stay the same, reducing neon's proportional ratio ($\frac{n_{\text{Ne}}}{n_{\text{tot}}}$).

Rubric Summary: 8 pts total: Directly targets Dalton's Law mechanics with a proportional conceptual check at the end. Independent step grading ensures partial errors don't penalize unrelated parts.